ASSOCIATED DETAIL SPECIFICATION

RESISTORS, FIXED, FILM, PRECISION,
ESTABLISHED RELIABILITY, STYLE RFP06

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for established reliability, precision, film, fixed resistors. Resistors are available within the full range of reactance, tolerance, and failure rates.

2. APPLICABLE DOCUMENTS

2.1 Government specification. Unless otherwise specified, the following specification, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-122 - Resistors, Fixed, Precision, Established Reliability, General Specification For.

(Copies of the specification required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-122, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Temperature characteristic. Resistors are available in characteristics A, B, C, D, E, and F.

3.3 Power rating. The power rating shall be .10 watt at 85°C (see 6.2). The applicable power rating curve is curve C.

3.4 Voltage rating. The continuous working voltage shall not exceed 200 volts.

3.5 Resistance. Minimum and maximum resistance values shall be as follows:

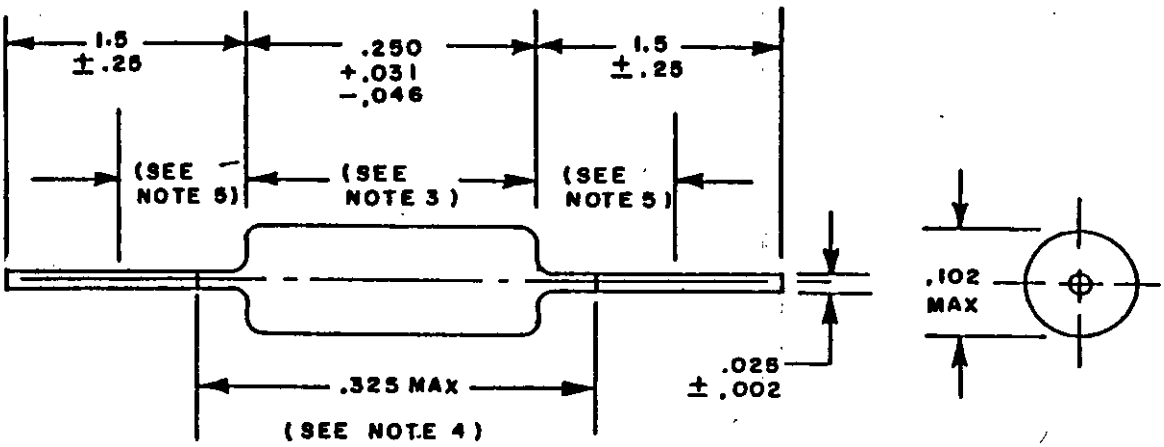
a. Minimum resistance - 10 ohms.

b. Maximum resistance - .5 megohm.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Space and Naval Warfare Systems Command, ATTN: SPAWAR 003-112, Department of the Navy, Washington, DC 20363-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at end of this document or by letter.

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Inches	mm	Inches	mm
.002	0.05	.125	3.18
.025	0.64	.25	6.35
.031	0.79	.325	8.26
.046	1.17	.5	12.7
.0625	1.588	1.5	38.1
.102	2.59		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 250 percent of the normal lead diameter.
4. Maximum length is "clean lead to clean lead".
5. Resistance measuring point shall be .5 ±.125 inch for resistance values of 10Ω or more and .0625 ±.025 inch for resistance values less than 10Ω.

FIGURE 1. Fixed resistors, precision.

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3.6 Weight. The maximum weight shall not exceed .00221 pound (1 gram).

3.7 Overload. Apply 6.25 times rated power to the resistor for 5 seconds. When tested as specified, the applied voltage shall not exceed 500 volts.

3.8 Dielectric withstanding voltage. The applicable test voltages are as follows:

a. Atmospheric pressure - 450 volts.

b. Barometric pressure (reduced) - 200 volts.

3.9 Marking. Due to size limitations, this style resistor shall be marked with the following minimum information:

042AM - Date code, reactance, resistance tolerance and failure rate
4775 - Resistance value and temperature characteristic
A75

The complete marking is required on the unit package. Where manufacturers are able to provide more information, the following is preferred in the sequence presented: Style, manufacturer's trademark, production lot code, and source code.

* 3.10 Thermal EMF. The thermal EMF shall not exceed 5.0 microvolts per degree Celsius.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-R-122.

5. PACKAGING

5.1 Packaging. Packaging requirements shall be in accordance with MIL-R-122.

6. NOTES

6.1 Notes. The notes specified in MIL-R-122 are applicable to this specification.

6.2 Power rating at 25°C. The power rating at 25°C for 2,000 hours of life test has been established at .2 watt, 200 volts, maximum voltage. It should be noted that the failure rate level is established at the single condition of 85°C, .10 watt and 10,000 hour life test duration.

6.4 Changes from previous issue. The margin of this specification is marked with an asterisk to indicate where a change from the previous issue was made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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